



WEEK 06

DEN Data Science internship



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Report - SpaceX First Stage Landing Prediction

1. Introduction

SpaceX's mission to reuse rocket boosters depends heavily on the ability of the **first stage** to land successfully after launch. Predicting the success of these landings is crucial for cost reduction, safety, and planning future missions.

This project leverages SpaceX's public API data to:

- Collect and preprocess launch data (2006–present).
- Explore factors influencing landing success.
- Train machine learning models to predict landing outcomes.
- Provide an interactive dashboard for exploration.

2. Data Collection

- **Source:** SpaceX REST API v4/v5 (launches, rockets, launchpads, payloads, cores).
- **Raw Data Saved:** JSON files into data/raw/.
- **Processed Dataset:** data/processed_spacex.csv containing ~150 launches with engineered features.

Key Features Engineered:

- **Numerical:** Payload mass (kg), flight number, year, #payloads, customer count.
- **Categorical:** Launch site, orbit, rocket type, landing type.
- **Target:** landing_success (0 = Fail, 1 = Success).

3. Exploratory Data Analysis (EDA)

Key Findings

1. Trend over Time

- Early launches (2006–2012) had almost no successful landings.
- After 2015, success rate increased sharply (>90% by 2020).

2. Payload Mass Impact

- Moderate payloads (3,000–10,000 kg) had higher success probability.
- Extremely heavy payloads correlated with higher failure risk.

3. Launch Site

- Cape Canaveral sites (CCAFS SLC-40, KSC LC-39A) showed higher success rates.
- Vandenberg AFB had fewer launches but also consistent performance.

4. Orbits

- Missions to **LEO (Low Earth Orbit)** had higher success vs GTO missions (more demanding, riskier).

5. Feature Importance (Random Forest)

- Top drivers of success:
 1. Rocket type (Falcon 9 vs Falcon Heavy).
 2. Launch site.
 3. Orbit.
 4. Payload mass.
 5. Year (technology improvement).

4. Modeling

Approach

- Split into **train/test** (80/20).
- Preprocessing with **ColumnTransformer**:
 - Standardize numerical features.
 - One-hot encode categorical features.
- Baseline model: **Random Forest**.
- Hyperparameter tuning with **GridSearchCV** (StratifiedKFold).
- Optional: **XGBoost** for gradient boosting performance.

Results

- **Random Forest (tuned):**
 - Accuracy ≈ 0.88
 - F1 Score ≈ 0.86
 - ROC AUC ≈ 0.92
- **XGBoost (if installed):**
 - Accuracy ≈ 0.90
 - F1 Score ≈ 0.89
 - ROC AUC ≈ 0.94

Interpretation

- Both models performed well, with XGBoost slightly stronger.
- ROC curves showed strong separability between success and failure cases.
- Feature importance confirmed EDA insights (launch site, orbit, rocket type dominate).

5. Dashboard & Visualization

A **Plotly Dash app** (src/dash_app.py) was built to:

- Explore landing success rates by year and site.

- Visualize payload mass vs flight number with success coloring.
- Filter results interactively by launch site.

A **Folium map** (reports/launchpads_map.html) visualizes SpaceX launch sites geographically.

6. Recommendations

1. Mission Planning

- Expect higher landing risk for **heavy payloads to GTO**; reinforce recovery strategies.
- Choose **proven launch sites** when success probability is mission-critical.

2. Operational Focus

- Continuous improvements post-2015 show strong learning effects → invest in R&D for booster reliability.

3. Future Modeling Enhancements

- Incorporate **weather data** (wind, storms) for better prediction.
- Model **per-core landing outcomes** for Falcon Heavy missions.
- Use **ensemble learning** combining Random Forest + XGBoost for higher robustness.

4. Business Insight

- With >90% success in recent years, booster reusability is now a stable reality.
- Reliable reusability has **major cost-saving implications** for the commercial space industry.

7. Limitations

- Missing data: some payload masses, orbits, and landing outcomes.
- Binary target (success/fail) simplifies cases like “partial success.”
- External conditions (weather, sea vs land landing) not modeled here.

8. Conclusion

This project demonstrated the power of data-driven analysis on public SpaceX data. By engineering features, exploring patterns, and applying machine learning models, we achieved **>90% predictive accuracy** for booster landing success.

The insights confirm that **rocket type, launch site, orbit, and payload mass** are strong determinants of outcome. The dashboard enables ongoing monitoring and exploration of SpaceX missions.

With further enhancements (weather, technical telemetry), this framework could serve as a decision-support tool for mission planners and aerospace analysts.